

4 The photograph shows yeast cells.



(Source: © Science Photo Library / Alamy Stock Photo)

In suitable conditions yeast cells respire and multiply.

A student formed the following hypothesis:

'the higher the temperature, the faster the rate of respiration'

Plan an investigation to test this hypothesis.

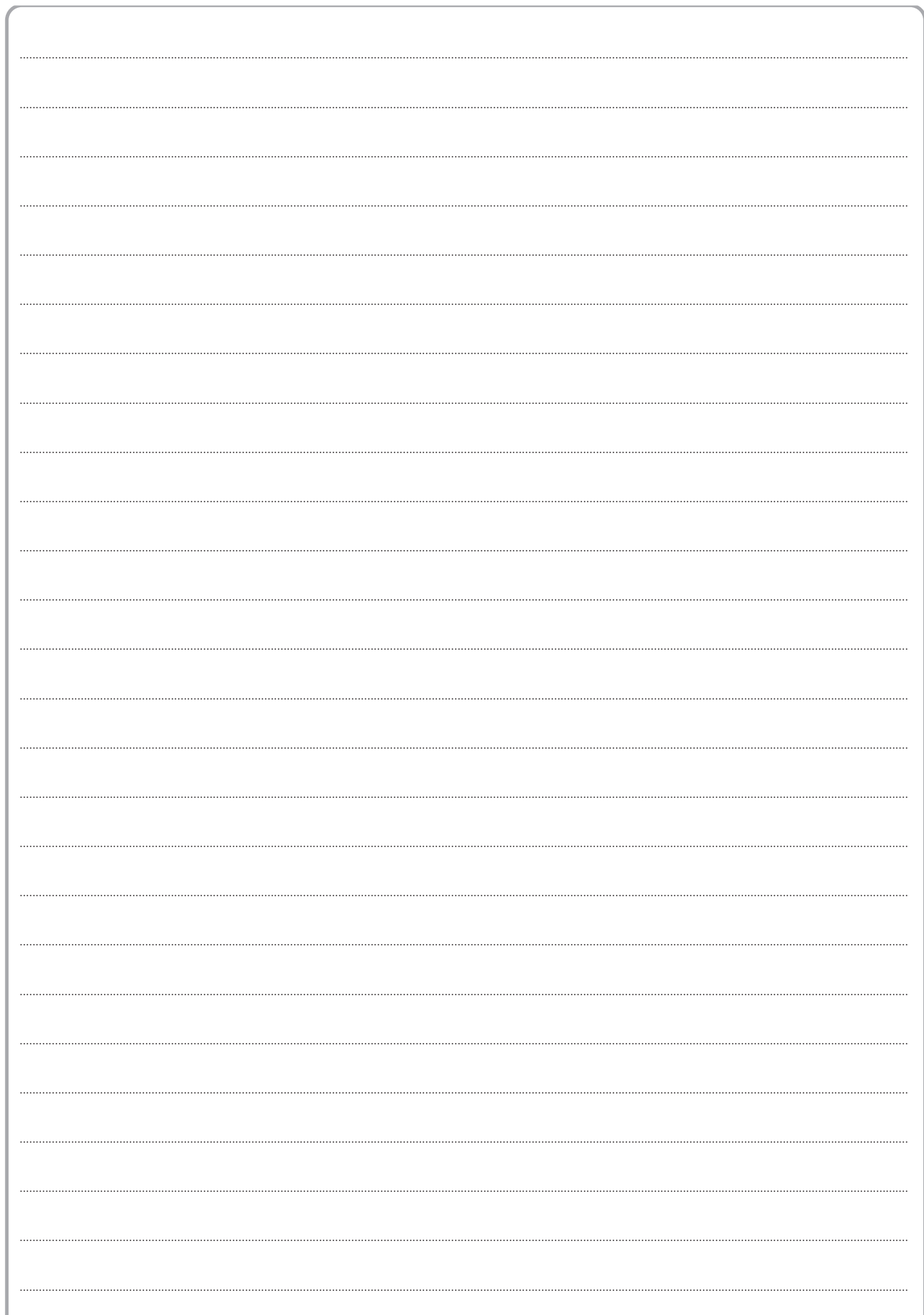
Your answer should give details under the following headings.

(a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(2)

(b) Devise a detailed method, including how you would control and monitor important variables.

(8)



(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

(d) Give **two** limitations of your proposed method.

(2)

(Total for Question 4 = 15 marks)

TOTAL FOR PAPER = 50 MARKS